

assignment-01

August 8, 2026

1 Assignment I

1.1 Import External Libraries

```
[1]: import numpy as np
```

1.2 5. Vector Inner Product Using Python

```
[2]: v1 = np.array([[np.sqrt(3)], [1]])  
v2 = np.array([[1], [1]])  
inner_product = np.vdot(v1, v2)  
  
print(f"Inner product: {inner_product:.3f}")  
print("v1 Shape:", v1.shape)  
print("v2 Shape:", v2.shape)
```

```
Inner product: 2.732  
v1 Shape: (2, 1)  
v2 Shape: (2, 1)
```

1.3 6. Inner Product with Complex Coefficient 1

```
[3]: v1 = np.array([[1j], [1]])  
v2 = np.array([[-1j], [1]])  
inner_result = np.vdot(v1, v2)  
  
print(f"Inner product: {inner_result:.3f}")
```

```
Inner product: 0.000+0.000j
```

1.4 7. Inner Product with Complex Coefficient 2

```
[4]: v1 = np.array([[1 + 2j], [1]])  
v2 = np.array([[1 - 1j], [1]])  
inner_result = np.vdot(v1, v2)  
  
print(f"Inner product: {inner_result:.3f}")
```

Inner product: $0.000-3.000j$
